Time-domain analysis of EEG event-related potentials (ERPs)

TUTORIAL IIA

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Synopsis

Event-related potentials (ERPs) are the measured brain response to a specific sensory, cognitive, or motor events. In the field of cognitive neuroscience and experimental psychology, the concept of ERPs has a longstanding tradition [1]; ERP-based assessment of brain mechanisms is still used today in many research domains, including research on perceptual processes, stimulus evaluation, decision making, and action execution. Meanwhile, ERPs have also been shown useful in the domain of brain-computer interfaces, such as in P300-speller applications and applications using the error-related potential (ErrP). ErrPs will be the focus of this tutorial, but all methods introduced here are general enough to be applicable to any other type of ERP as well.

In the first part of tutorial II you will learn about time-domain analyses of ERPs. Please note that approximately 90% of the research papers on ERPs do not go beyond time-domain analysis, which is why these methods comprise the basic fundament of ERP analysis.

The goal of this tutorial is to work out possible neurophysiological effects in the given EEG dataset that manifest in time-domain ERP. Importantly, you will do this in an exploratory fashion, that is, without any prior assumptions about possible underlying effects. To this end you will demonstrate if and to what extent these effects are valid (statistically significant).

Homework submission will contain the following items:

- One report containing answers to all questions (from part A, B, and C) placed in the tutorial instructions.
- Your commented code (separated in part A, B and C)

The EEG dataset

The given EEG dataset comprises data from several subjects who performed the delayed cursor task developed at the Chair for Cognitive Systems [2]. Details about the experimental protocol can be obtained from the lecture slides. The table below shows an overview of information about how the dataset was acquired; the descriptions below explain the structure of the data.

EEG setup	Brain Products actiChamp amplifier with 32 active gel-based electrodes (27 EEG, 3 EOG, and 2 reference channels)
Referencing	Average mastoids (TP9+TP10)
Sampling rate	256 Hz (downsampled from original 1000 Hz)
Number of subjects	17
Number of trials per subject	150
Total number of trials	2550
Number of conditions	2 (non-error, error)
Number of trials per condition	~65% non-error and ~35% error trials

Data preprocessing

The following steps of preprocessing and data segmentation were performed:

1. Bandpass filtering of raw EEG data with cutoff frequencies of 1-40 Hz.
2. Identification of artifact-contaminated EEG channels; channel rejection and interpolation using neighboring channels.
3. Reduction of EOG artifact using Schloegl's regression method [3]; removal of EOG channels.
4. Rereferencing to common average reference (CAR).
5. Downsampling to 256 Hz.
6. Data segmentation / epoching: extracting data segments of -500 to 1500 ms time-locked to the moment of feedback presentation (movement of the cursor).

Dataset structure

The data is organized in a matlab struct with the following content:

.data	4D-matrix containing the segmented EEG data (27 channels * 512 timepoints * 150 trials * 17 subjects)
.chanlocs	Contains both the channel labels and channel location in two different coordinate systems (needed for topographic visualization)

.times	A vector associating sample-count with time-information (-500 to 1500 ms)
.subjects	A cell-array containing the subject IDs
.srate	The sampling rate
.labelcodes	A cell-array containing the meaning of the numerical labels in ***.label***
.labels	A 2D-array containing the labels of all trials and subjects
.unit	The physical unit of the data (µV – microvolts)
.chanlist	A list of channel-labels (complementary to ***.chanlocs***)

1: Exploring the dataset

Load the dataset into your Matlab workspace and explore its content (compare with the table above). Make yourself familiar with the data and discuss with your classmates and ask if things are unclear.

2: Single-channel baseline correction

Despite high-pass filtering with a cutoff frequency of 1 Hz, single trials may still have a DC-offset. This offset leaves unnecessary noise in our data which can have a negative effect in trial-averaging (see Section 3). Therefore single-trial ERP time-courses are usually *baseline corrected* before being averaged.

Baseline correct your data by subtracting the mean of time-period -200 to 0 ms from the entire trial. Do this for all channels, trials and subjects individually.

3a: Plotting some data

Start out by plotting a single trial ERP time-course from one channel, one trial, and one subject and using the Matlab `plot` function. Use the ***.time*** vector to properly scale your x-axis.

Now, superimpose two single trial ERP time-courses (use the Matlab `hold on` command), one from each condition: non-error and error. Keep the channel and subject fixed for this comparison, e.g. select one trial from condition non-error and one from condition error from the same subject and channel.

Q-A1 (5 points): Can you observe any qualitative differences between the two conditions based on the comparison of these two trials? Please describe in your own words. Report

which channel and subject you have chosen and put the image with appropriate axes labels in your report.

3b: Visualizing single-channel grand-average ERP time-courses

Due to high intertrial-variance, ERP effects are usually only visible after averaging across many trials of the same condition. We will focus here on channel Cz where we expect the strongest effect.

Start out by focusing on one subject and compute the average ERP of channel Cz for each of the two conditions: non-error and error. This leaves you with two time-courses. Now compute the *difference average* by subtracting the non-error time-course from the error time-course. Plot all three time-courses in a single Matlab plot using different colors (blue: non-error, red: error, black: difference).

Repeat the procedure for all 17 subjects and compute the final *grand average* by averaging the single-subject time courses of each of the conditions and the condition difference.

Q-A2 (10 points): Describe what you observe in the grand average time-course of channel Cz with your own words and put the plot in your report. At which time points are the conditions different? Can these differences be associated with well-known ERP-components?

4: Creating an ERP-image (BONUS)

From the previous analysis we may conclude that there are distinct differences between the ERPs of both conditions. However, these differences may be driven by a few trials with very strong deflections. By looking at the average (or grand average) only, we cannot determine whether we see a consistent effect across trials. A useful method to double-check is to create ERP-images. These visualize single-trial ERP time-courses in a stacked fashion and show if throughs and peaks are consistent across trials.

Focus on one subject (s01) and channel (Cz) and create a 2D plot (timepoints*trials). Use the Matlab function `contourf` with an interpolation factor of 20 and remove the lines (`'linecolor','none'`). Sort the trials into two matrices: non-error and error and plot them in two separate contour-plots. In order to see the effects more clearly it is recommended to smooth your data-matrices with a 2D gaussian kernel before visualizing. Use Matlab function `imgaussfilt` and choose an appropriate standard deviation.

Q-A3 (10 points - BONUS): Describe what you observe in the ERP images and place them in your report. Are the effects (peaks and throughs) of the (grand) average ERP consistent

across trials? Plot the ERP images of a few other subjects and compare them – are the images different among subjects? Describe your observations.

5: Visualizing ERPs in topographic activation maps

The previous investigation helped us qualitatively understand the ERP as it evolves in time. However, it does not tell us anything about how the ERP evolves in space, e.g. at which electrodes the ERP is most prominent in time. Topographic activation maps help us to look deeper into this.

Start out by choosing a specific time point of interest, e.g. around the early negative deflection of the condition difference. Use the function `topoplot` to visualize the topographic activation map at the time point you selected. `Topoplot` receives a vector of length "number of channels" + the channel location information given in the provided dataset.

Visualize the grand average topographic activation maps of both conditions and the condition difference in a single Matlab figure (use Matlab function `subplot`).

Q-A4 (10 points): Describe what you observe in the grand average topographic activation maps for the selected time-point. Where are the most prominent activations located? Use proper terminology to describe the locations (frontal, central, temporal, parietal, occipital sites).

A-6: ROI-based statistical analysis across subjects

So far, we have only qualitatively analyzed our data. The last two analyses will allow us to make quantitative statements about the effects we have observed in the data.

We will focus first on a region-of-interest (ROI)-based statistical analysis: ErrPs are understood to have two prominent components, the N2 (or N200) and the P3 (or P300) component [4]. These components are well-defined in specific time-periods [5]:

- N2 -> negative deflection in period 180-325 ms over central (and frontal) sites
- P3 -> positive deflection in period 300-600 ms over central (and parietal) sites

Pool (average) the data of each subject's difference average in pre-defined ROIs: average in time using the above definitions for N2 and P3 and in space by selecting the following central channels (Fz, FC1, FC2, Cz, CP1, CP2, Pz).

This gives you two separate distributions, one per ROIs, each with one value per subject. Now, test these distributions for each ROI against zero by using the standard t-test (Matlab

function `ttest`) and a significance threshold of *p<0.05*. The null-hypothesis is that samples in the tested distribution come from a normal distribution with mean-value of zero (no effect).

Q-A6 (10 points): Formulate the null-hypothesis of the two tests you performed. Report the results of the two ROI-based statistical tests and report whether you accept or reject the null-hypotheses of both tests; report also the resulting p-values and the means of the distributions.

References / further reading:

[1] Luck, S. J. (2014). *An introduction to the event-related potential technique*. MIT press.
[2] Ehrlich, S. K., & Cheng, G. (2019). A Feasibility Study for Validating Robot Actions Using EEG-Based Error-Related Potentials. *International Journal of Social Robotics*, *11*(2), 271-283.
[3] Schlögl, A., Keinrath, C., Zimmermann, D., Scherer, R., Leeb, R., & Pfurtscheller, G. (2007). A fully automated correction method of EOG artifacts in EEG recordings. *Clinical neurophysiology*, *118*(1), 98-104.
[4] Chavarriaga, R., Sobolewski, A., & Millán, J. D. R. (2014). Errare machinale est: the use of error-related potentials in brain-machine interfaces. *Frontiers in neuroscience*, *8*, 208.
[5] Patel, S. H., & Azzam, P. N. (2005). Characterization of N200 and P300: selected studies of the event-related potential. *International journal of medical sciences*, *2*(4), 147.